Natural Language Processing Coursework

Detecting Patronising and Condescending Language (PCL)

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Exercise 1: Critical Paper Review

Q1. State the primary contributions of this work.
The authors make several foundational contributions to the computational study of Patronizing and Condescending Language (PCL). Primarily, they introduce the novel "Don't Patronize Me!" dataset, comprising 10,637 annotated paragraphs extracted from news stories across 20 English-speaking countries. To support this dataset, they propose a two-level taxonomy for PCL, grouping linguistic traits into three high-level categories: *The Saviour*, *The Expert*, and *The Poet*. Finally, they formalise PCL detection as an NLP task and establish baseline performances using models ranging from standard Support Vector Machines to fine-tuned pre-trained Language Models like BERT and RoBERTa.

Q2. Identify the technical strengths that justify the paper's publication.
The paper's primary strength lies in its rigorous and self-aware methodology. Recognising the inherent subjectivity in identifying condescension, the authors implemented a robust two-step annotation process utilising three expert annotators. They employed a 5-point scale specifically designed to capture partial disagreements and nuance in borderline cases. Furthermore, their baseline evaluation is comprehensive, testing a wide array of architectures (SVM-WV, SVM-BOW, BiLSTM, and multiple BERT variants) to definitively prove that PCL detection is feasible and that contextual sequence classifiers significantly outperform random guessing. The paper concludes with a highly perceptive error analysis, successfully diagnosing that even advanced models struggle with figurative language (metaphors) and instances requiring external situational knowledge.

Q3. Highlight the key weaknesses or areas where the authors failed to provide sufficient evidence.
While foundational, the study contains notable limitations. The initial paragraph-level Inter-Annotator Agreement (IAA) was relatively low at 41%, which, while reflecting the subjective nature of the task, limits the objective reliability of the dataset's ground truth. Furthermore, there is a severe data sparsity issue for certain PCL subcategories; for example, the "The poorer, the merrier" category contains only 64 total samples, directly contributing to the baseline models' failure to correctly classify it. Lastly, while the authors correctly conclude that accurately modelling PCL requires external "knowledge of the world and common sense," their tested baselines rely purely on standard sequence representations and lack any explicit mechanisms to inject or retrieve this required external knowledge.

Exercise 2: Exploratory Data Analysis (EDA)

Technique 1: Class Distribution Analysis

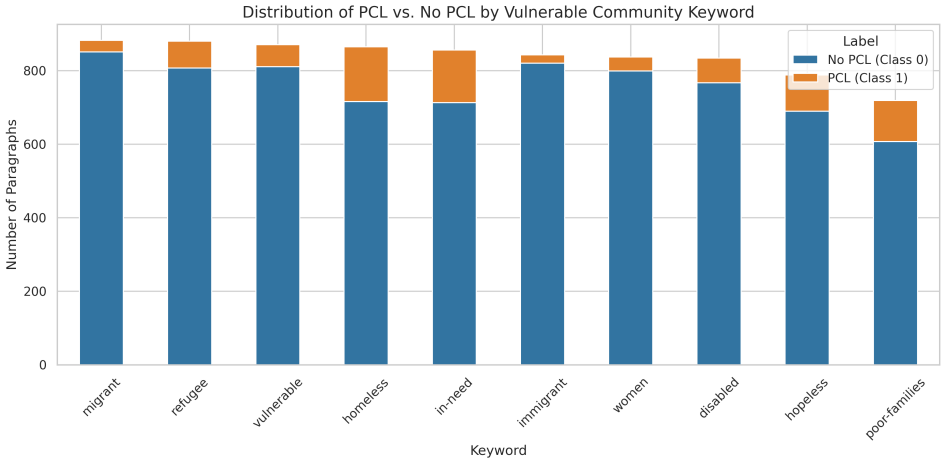

Figure 1: Distribution of PCL vs. No PCL, categorised by vulnerable community Keyword.

Analysis:
The dataset exhibits a severe overall class imbalance, with the positive PCL class representing less than 10% of the total data. Furthermore, visual analysis demonstrates that this imbalance is highly conditional upon the targeted vulnerable community. Paragraphs containing keywords such as 'homeless', 'in-need', and 'poor-families' contain a visibly higher proportion of PCL instances compared to keywords like 'migrant', 'immigrant', and 'women', which are overwhelmingly negative.

Impact Statement:
This conditional imbalance will naturally bias a standard cross-entropy loss function towards the majority 'No PCL' class, risking a low F1 score on the positive class. To mitigate this, my training pipeline must incorporate robust class weighting. Additionally, to resolve the sheer lack of positive examples, I will implement targeted data augmentation on the minority class to artificially balance the distribution before training.

Technique 2: Linguistic Analysis through Part-of-Speech (POS) Tagging

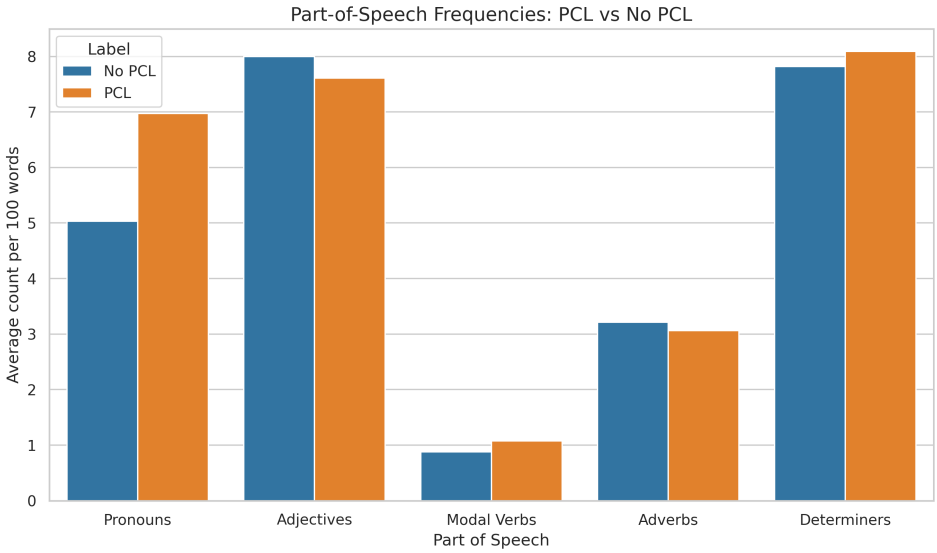

Figure 2: Average Part-of-Speech frequency per 100 words, comparing PCL and No PCL paragraphs.

Analysis:
An analysis of Part-of-Speech frequencies reveals that PCL paragraphs exhibit a nearly 40% higher density of pronouns compared to standard news text, alongside slight increases in determiners and modal verbs. Conversely, standard descriptive markers like adjectives and adverbs are actually slightly lower in PCL contexts. This supports the original paper's sociolinguistic claim that PCL relies heavily on structural "distancing" language (separating the 'us' from the 'them') and prescriptive 'expert' tones, rather than relying solely on flowery vocabulary.

Impact Statement:
Because simple lexical markers do not clearly isolate condescension, shallow models (like TF-IDF) are insufficient. The heavy reliance on pronoun distancing and subtle modal structures requires a model capable of deep coreference resolution and long-range syntactic understanding. This justifies bypassing smaller baselines entirely and utilising a large contextual Transformer architecture, specifically RoBERTa-large, coupled with prompt-based fine-tuning to better capture semantic nuance.

Exercise 3: Describe your proposed approach

Proposed Approach
To surpass the RoBERTa-base baseline, I propose a 5-fold ensemble architecture utilising RoBERTa-large, augmented by Prompt-Based Learning (Masked Token Prediction), context-aware data augmentation, and dampened inverse class weighting.

Rather than using a standard classification head, the task is reformulated as a masked language modelling problem. The model is fed the original text appended with a prompt: *"Question: Is the author being patronizing or condescending towards {keyword}? Answer: [MASK]"*. The model is then fine-tuned to predict the probabilities of the tokens "Yes" (Class 1) and "No" (Class 0) at the masked position.

To address the severe class imbalance identified in the EDA, I utilize the 'TextAttack' library to artificially augment the minority PCL class. Crucially, I employ the 'spaCy' NER model during this augmentation to freeze and protect specific entity tags (GPE, LOC, NORP) alongside the target keyword. This ensures that the socio-political context, which is vital for PCL detection, is preserved while synthetically expanding the minority class, successfully shifting the training ratio from roughly 1:9 to 1:4.

During training, I implement a Dampened Inverse Class Weighting mechanism within the Cross-Entropy loss to penalize false negatives without destabilizing the gradients. The weight for class c is calculated as:

$$w_c = 1.0 + \alpha \left(\frac{N}{2 \times N_c} - 1.0 \right)$$

where N is the total number of samples, N_c is the sample count for class c, and α is a dampening hyperparameter tuned via Optuna, alongside learning rate and weight decay. Implementing a dampened class weighting over a full class weighting prevents gradient explosion and allows Optuna to find a value that boosts the F1 with class weighting in this way. Finally, the model is trained across a 5-fold Stratified Cross-Validation split. During inference, the output probabilities from all five distinctly trained models are averaged to produce a soft-voted final prediction.

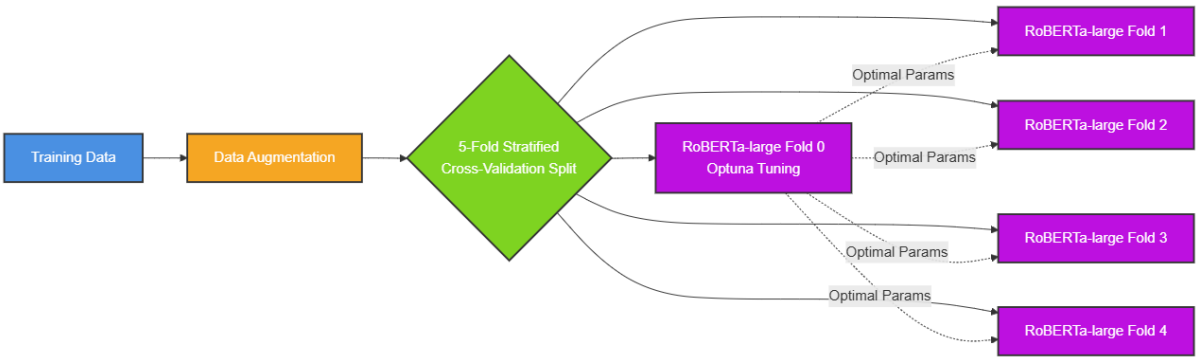

Figure 3: Proposed training pipeline.

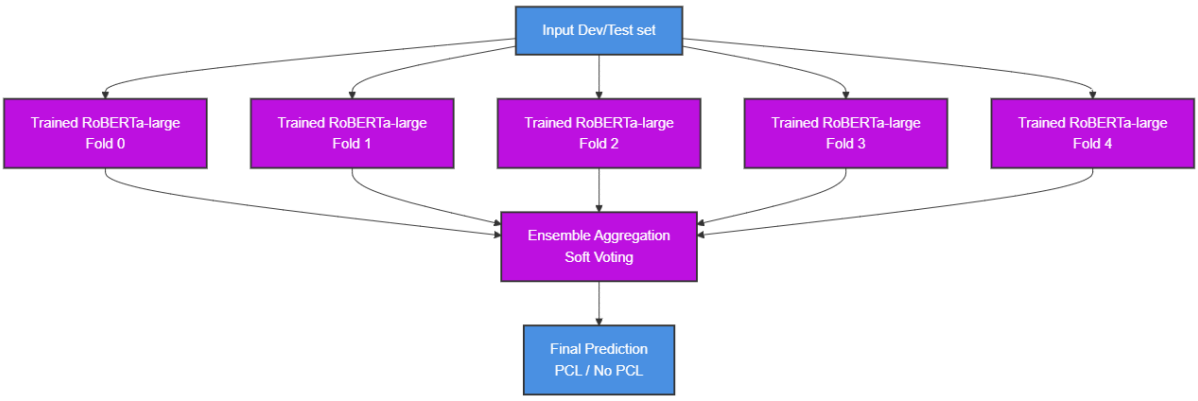

Figure 4: Proposed inference pipeline. This just uses the 5 model ensemble.

Rationale & Expected Outcome
This approach directly addresses the specific sociolinguistic and structural challenges of PCL. By utilising prompt-based learning, the gap between RoBERTa-large's pre-training objective and the downstream task is bridged, allowing the model to leverage its native linguistic reasoning better than a randomly initialised classification head. Furthermore, the NER-aware data augmentation ensures that the model learns the syntactic structure of condescension (the "distancing" pronouns identified in the EDA) without hallucinating or losing vital geographical context.

I expect this framework to significantly outperform the RoBERTa-base baseline in terms of the positive class F1-score. By tackling the class imbalance from two distinct angles (synthetic oversampling and mathematical loss weighting) and utilising a 5-fold ensemble to smooth out variance, the model will be far more resilient to the subtleties of PCL and less prone to overfitting on the majority class.

Exercise 5: Local Evaluation

Ablation Study and Global Performance

To validate the effectiveness of the proposed methodology, I conducted an ablation study comparing three variations of the architecture on the internal development set. The results demonstrate that ensembling and model selection played a critical role in final performance.

Model Variation	Architecture	Optimal Threshold	Dev Set F1 (Positive)
Single Fold	RoBERTa-large	0.42	0.6216
5-Fold Ensemble	**RoBERTa-large**	**0.46**	**0.6250**
5-Fold Ensemble	DeBERTa-v3-large	0.30	0.5249

Table 1: Ablation study evaluating the impact of ensembling and base model selection.

The optimal hyperparameters that Optuna found for the RoBERTa-large 5-fold ensemble

were:

$$\alpha = 0.5790$$
$$\text{lr} = 5.245 \times 10^{-6}$$
$$\text{weight_decay} = 0.03716$$

The RoBERTa-large 5-fold ensemble achieved the highest Dev Set F1 score of 0.6250, proving that averaging logits across differently validated folds effectively smooths out edge-case misclassifications. Interestingly, despite DeBERTa-v3-large generally outperforming RoBERTa on standard benchmark leaderboards, it severely underperformed here. This suggests that RoBERTa's design may be more suited to my approach as it was trained with masked token modelling, whereas DeBERTa-v3 was trained using Replaced Token Detection (RTD) via an ELECTRA-style discriminator. Because my prompt-based approach relies directly on filling a [MASK] token, RoBERTa's native pre-training objective aligns perfectly with the fine-tuning task, granting it a distinct architectural advantage over DeBERTa.

Furthermore, analysing the optimal decision thresholds provides additional insight into model calibration. In severely imbalanced datasets, models often exhibit low confidence when predicting the minority class, requiring the classification threshold to be drastically lowered to capture true positives. This is evident in the DeBERTa-v3-large ensemble, which required an optimal threshold of 0.30, indicating poor calibration and an overall hesitation to predict PCL. In contrast, the RoBERTa-large ensemble achieved its maximum F1 score at a threshold of 0.46. Remaining this close to the standard 0.50 decision boundary demonstrates that the RoBERTa ensemble is significantly better calibrated; its output probabilities naturally separate the classes without requiring an aggressive post-hoc shift to compensate for the minority class.

Keyword	**Fold 0 Support (Class 0)**	**Fold 0 Support (Class 1)**	**Dev Set F1 Score**
in-need	588	119	0.80
vulnerable	629	48	0.71
migrant	687	25	0.67
hopeless	545	74	0.64
homeless	580	120	0.63
poor-families	475	95	0.56
refugee	649	59	0.53
disabled	621	48	0.46
women	633	31	0.45
immigrant	658	16	0.22

Table 2: Breakdown of F1 performance by keyword for the best model (RoBERTa-large Ensemble), mapped against the training support from a representative fold (Fold 0) after augmentation. *Note: Whilst the supports are only shown for Fold 0 excluding the validation set, the supports for the other folds are roughly the same.*

When evaluating the best model's performance on a per-keyword basis (Table 2), a complex relationship emerges between raw class distribution and model performance. At the extremes, data volume is a clear predictor: the model excels at detecting PCL for "in-need" (119 Class 1 instances in Fold 0; 0.80 F1), which enjoys a healthy baseline of positive training examples. Conversely, it struggles massively with "immigrant" (only 16 positive instances; 0.22 F1), where severe data starvation meant my TextAttack augmentation pipeline had too few diverse, high-quality root sentences to effectively extrapolate from.
However, the correlation between raw data volume and F1 score is not absolute. For example, "migrant" and "vulnerable" achieve strong F1 scores (0.67 and 0.71, respectively) despite having

very low original positive support (25 and 48 instances). Conversely, "homeless" possesses the highest number of positive training examples (120) but only yields a moderate F1 score of 0.63. This discrepancy suggests that the model's success depends heavily on the *consistency* and *unambiguity* of the condescending language used in the media. Keywords that trigger highly repetitive, distinct "Saviour" or "Compassion" narratives are easier for the prompt-based model to learn even with sparse data. In contrast, groups subjected to highly varied, politicised, or context-dependent discourse (such as "homeless" or "women") make the PCL boundary much blurrier, requiring a more sophisticated understanding of pragmatics than the model currently possesses.

Error Analysis

While the global F1 score is strong, manual inspection of the model's most confident False Positives (FPs) and False Negatives (FNs) reveals distinct patterns in its logic and limitations.

To ground this qualitative analysis, it is useful to first examine the exact distribution of the model's errors. Table 3 presents the confusion matrix for the RoBERTa-large 5-fold ensemble on the development set.

Actual \ Predicted	**PCL (Class 1)**	**No PCL (Class 0)**
PCL (Class 1)	120 (True Positive)	79 (False Negative)
No PCL (Class 0)	65 (False Positive)	1830 (True Negative)

Table 3: Confusion matrix for the best model (RoBERTa-large 5-Fold Ensemble) on the Dev Set.

The matrix reveals a highly balanced error distribution. In many imbalanced classification tasks, models tend to severely skew towards either False Negatives (ignoring the minority class) or False Positives (over-predicting to compensate for loss weights). Here, the tight margin between missed PCL (79) and hallucinated PCL (65) demonstrates that the dampened inverse class weighting and ensemble thresholding successfully calibrated the model's decision boundary. However, to understand why these 144 combined errors occurred, we must manually inspect the failed predictions.

Analysis of False Negatives (Missed PCL):
The model frequently misses PCL when it is disguised as objective administrative reporting or lacks explicit "Saviour" vocabulary. For example, the model failed to flag: *"...sensitise poor families to hygienic conditions."* To a human reader, this is clearly condescending (The Expert / Presupposition), as it implicitly assumes poor families are inherently unhygienic. However, because it is written in the sterile tone of an NGO report and lacks flowery adjectives or pitying pronouns, the model's heuristic fails to trigger. Similarly, the model missed: *"...20 disabled undergraduate students... receiving the same salaries as able-bodied colleagues."* The model treats this as an objective employment statistic, missing the underlying "Saviour" narrative that equal pay for disabled individuals is an extraordinary act worthy of news coverage.

Analysis of False Positives (Hallucinated PCL):
Conversely, the model is easily tricked into predicting PCL when it encounters highly emotive language or metaphors, even when the context does not target a vulnerable group. For instance, the model confidently flagged: *"Not homeless – but STARVING for success."* This is clearly a metaphor used to describe ambition, not a literal condescending remark about the homeless community. Yet, the combination of the keyword "homeless" and dramatic capitalization

caused the model to hallucinate PCL. Similarly, the model flagged: *"It's the politicians... All we hear from them is negativity... feeling hopelessness."* Here, the model over-indexed on the word "hopelessness" without understanding that the text is a macro-political critique, rather than a condescending judgment of a specific vulnerable group.

Ultimately, this error analysis confirms the original authors' hypothesis: while syntax and vocabulary can identify the majority of PCL, achieving state-of-the-art performance requires a model capable of genuine semantic world-knowledge and contextual pragmatics. If I were to attempt to improve on my approach, I would do Task Adaptive Pre-training (TAP) on world-knowledge to address this issue before fine tuning on the dataset.